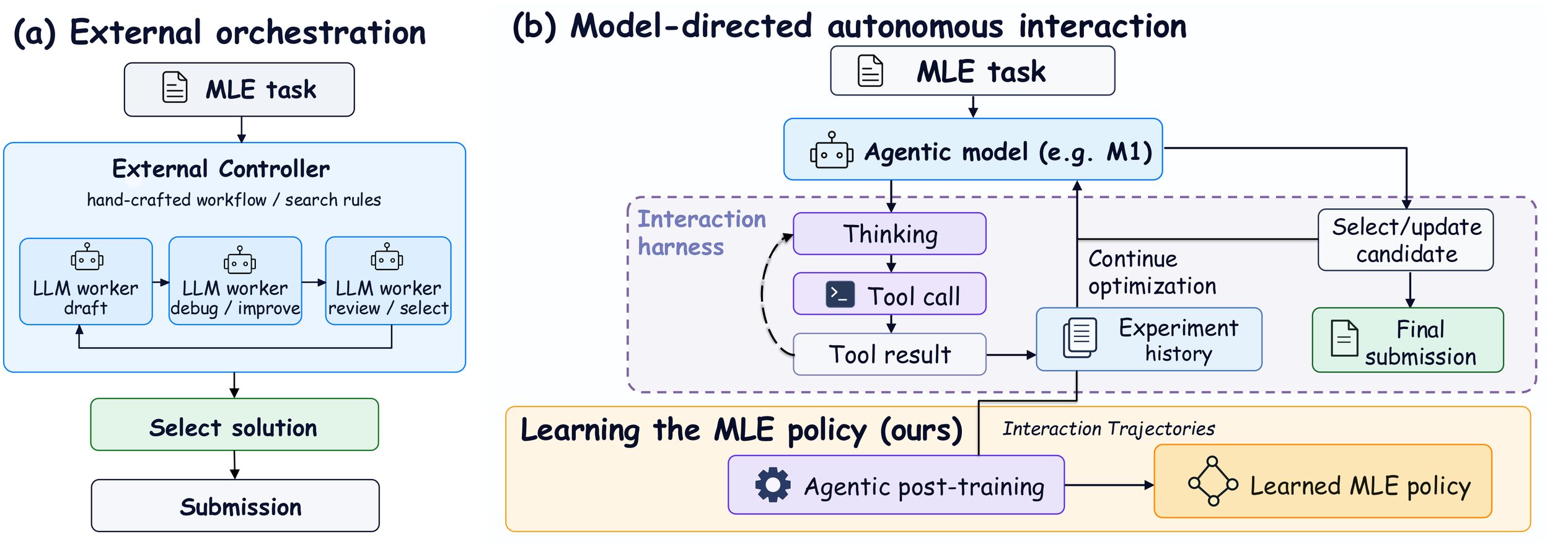

(a) External orchestration
MLE task
External Controller
hand-crafted workflow / search rules
LLM worker draft
LLM worker debug / improve
LLM worker review / select
Select solution
Submission
(b) Model-directed autonomous interaction
MLE task
Agentic model (e.g. M1)
Interaction harness
Thinking
Tool call
Tool result
Experiment history
Continue optimization
Select/update candidate
Final submission
Learning the MLE policy (ours)
Interaction Trajectories
Agentic post-training
Learned MLE policy